

Experiment 10

1. Aim

Analyse System logs to detect suspicious activities and generate audit report using tools like auditd / journalctl.

2. Hardware / Software Required

- Linux operating system (Ubuntu/CentOS/Kali Linux)
- auditd – Linux auditing framework
- journalctl – Systemd journal log analysis utility
- Terminal/Command Line
- Text editor for preparing the audit report

3. Theory.

System logs record important activities performed by the operating system, users, applications, and services. Log analysis is an important part of system monitoring, incident detection, and digital forensics because suspicious activities often leave traces in system logs.

1. auditd

auditd (Linux Audit Daemon) records security-relevant events such as:

- User login and logout activities
- Failed authentication attempts
- Changes to files and directories
- Use of privileged commands
- Changes in user accounts
- Permission and ownership changes
- System configuration modifications

Audit logs are commonly stored in: **/var/log/audit/audit.log**

Useful commands include:

- `sudo systemctl status auditd`
- `sudo ausearch -m USER_LOGIN`
- `sudo ausearch -m USER_AUTH`
- `sudo aureport -au`

2. journalctl

journalctl is used to view and analyze logs collected by systemd-journald. It can be used to investigate system boots, services, authentication events, errors, and other system activities.

Examples:

- `sudo journalctl`
- `sudo journalctl -b`
- `sudo journalctl -p warning`
- `sudo journalctl --since "today"`
- `sudo journalctl -u ssh`

3. Detecting Suspicious Activities

During analysis, events that may indicate suspicious behavior are identified, such as:

- Multiple failed login attempts
- Unexpected successful logins
- Unauthorized privilege escalation
- Unknown users or accounts
- Repeated service failures
- Unexpected changes to sensitive files
- Unusual activity outside normal working periods

The identified events can be correlated with timestamps, usernames, source information, and affected services to determine whether they represent potential security incidents.

4. Audit Report

The final audit report should document:

- Date and time of the event
- Type of activity
- User/account involved
- Source or affected system/service
- Evidence from the logs
- Suspicion level
- Recommended action

Sr. No.	Purpose	Command
1	Check auditd status	sudo systemctl status auditd
2	Start auditd	sudo systemctl start auditd
3	Enable auditd at boot	sudo systemctl enable auditd
4	View audit logs	sudo cat /var/log/audit/audit.log
5	View latest audit entries	sudo tail -n 50 /var/log/audit/audit.log
6	Search user login events	sudo ausearch -m USER_LOGIN
7	Search authentication events	sudo ausearch -m USER_AUTH
8	Generate authentication report	sudo aureport -au
9	Generate login report	sudo aureport -l
10	View executed commands	sudo aureport -x
11	Generate complete audit report	sudo aureport
12	Monitor /etc/passwd changes	sudo auditctl -w /etc/passwd -p wa -k passwd_changes
13	Search /etc/passwd changes	sudo ausearch -k passwd_changes
14	Monitor /etc/shadow access	sudo auditctl -w /etc/shadow -p r -k shadow_access
15	Search /etc/shadow access	sudo ausearch -k shadow_access
16	View all system journal logs	sudo journalctl
17	View logs from current boot	sudo journalctl -b
18	View today's logs	sudo journalctl --since today
19	View logs from the last hour	sudo journalctl --since "1 hour ago"
20	View warning messages	sudo journalctl -p warning

21	View error messages	<code>sudo journalctl -p err</code>
22	View SSH logs	<code>sudo journalctl -u ssh</code>
23	Search failed activities	<code>sudo journalctl grep -i "failed"</code>
24	Check failed authentication (Ubuntu/Debian)	<code>sudo grep -i "failed" /var/log/auth.log</code>
25	Save audit report to a file	<code>sudo aureport > system_audit_report.txt</code>
26	Save login events to a file	<code>sudo ausearch -m USER_LOGIN > login_events.txt</code>

4. Procedure

1. Open the **Linux terminal** with sudo privileges.
2. Check the status of the **auditd** service.
3. Start auditd if it is not running.
4. View recent audit logs from `/var/log/audit/audit.log`.
5. Use ausearch to check **user login and authentication events**.
6. Use aureport to generate **login, authentication, and command-execution reports**.
7. Configure audit rules to monitor sensitive files such as `/etc/passwd`.
8. Perform a controlled file access/change and use ausearch to identify the recorded event.
9. Use journalctl to view **system and service logs**.
10. Filter journal logs for **warnings, errors, and failed activities**.
11. Examine SSH logs for suspicious or repeated login attempts.
12. Analyze the collected logs for **failed logins, unauthorized changes, privilege escalation, and unusual activities**.
13. Correlate suspicious events using **timestamps, usernames, services, and event types**.
14. Classify the identified events as **normal or suspicious**.
15. Generate and save the final **system audit report** containing the findings, evidence, and recommended actions.

5. Output

```
Session Actions Edit View Help
kali@kali: ~
└─$ sudo apt install auditd audispd-plugins -y
The following packages were automatically installed and are no longer required:
idle-python3.13 libllhttp9.4 libpython3.13-testsuite python3.13-dev python3.13-examples python3.13-minimal
libgit2-1.9 libpython3.13-dev python3.13 python3.13-doc python3.13-full python3.13-venv
Use 'sudo apt autoremove' to remove them.

Installing:
  audispd-plugins auditd

Installing dependencies:
  libauparse0t64 libauplugin1

Summary:
  Upgrading: 0, Installing: 4, Removing: 0, Not Upgrading: 1126
  Download size: 384 kB
  Space needed: 1,214 kB / 50.9 GB available

Get:1 http://http.kali.org/kali kali-rolling/main amd64 libauparse0t64 amd64 1:4.1.2-1+b1 [73.2 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 auditd amd64 1:4.1.2-1+b1 [235 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 libauplugin1 amd64 1:4.1.2-1+b1 [23.2 kB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 audispd-plugins amd64 1:4.1.2-1+b1 [51.8 kB]
Fetched 384 kB in 2s (227 kB/s)
Selecting previously unselected package libauparse0t64:amd64.
(Reading database... 444000 files and directories currently installed.)
Preparing to unpack .../libauparse0t64_1%3a4.1.2-1+b1_amd64.deb...
Adding "diversion of /lib/x86_64-linux-gnu/libauparse.so.0 to /lib/x86_64-linux-gnu/libauparse.so.0.usr-is-merged by libauparse0t64"
Adding "diversion of /lib/x86_64-linux-gnu/libauparse.so.0.0.0 to /lib/x86_64-linux-gnu/libauparse.so.0.0.0.usr-is-merged by libauparse0t64"
Unpacking libauparse0t64:amd64 (1:4.1.2-1+b1)...
Selecting previously unselected package libauplugin1:amd64.
Preparing to unpack .../libauplugin1_1%3a4.1.2-1+b1_amd64.deb...
Unpacking libauplugin1:amd64 (1:4.1.2-1+b1)...
Selecting previously unselected package auditd.
Preparing to unpack .../auditd_1%3a4.1.2-1+b1_amd64.deb...
Unpacking auditd (1:4.1.2-1+b1)...
Selecting previously unselected package audispd-plugins.
Preparing to unpack .../audispd-plugins_1%3a4.1.2-1+b1_amd64.deb...
```

```
Session Actions Edit View Help
kali@kali: ~
└─$ sudo systemctl status auditd
o auditd.service - Security Audit Logging Service
   Loaded: loaded (/usr/lib/systemd/system/auditd.service; disabled; preset: disabled)
   Active: inactive (dead)
   Docs: man:auditd(8)
         https://github.com/linux-audit/audit-documentation

(kali㉿kali)-[~]
└─$ sudo systemctl start auditd

(kali㉿kali)-[~]
└─$ sudo systemctl enable auditd
Created symlink '/etc/systemd/system/multi-user.target.wants/auditd.service' → '/usr/lib/systemd/system/auditd.service'.

(kali㉿kali)-[~]
└─$ sudo cat /var/log/audit/audit.log
type=DAEMON_START msg=audit(1788841332.181:931): op=start ver=4.1.2 format=enriched kernel=6.19.14+kali-amd64 auid=4294967295 pid=3758 uid=0 ses=
4294967295 subj=unconfined res=successAUID="unset" UID="root"
type=SERVICE_START msg=audit(1788841332.180:3): pid=1 uid=0 auid=4294967295 ses=4294967295 subj=unconfined msg='unit=auditd comm="systemd" exe="/
usr/lib/systemd/systemd" hostname=? addr=? terminal=? res=success'UID="root" AUID="unset"
type=USER_END msg=audit(1788841332.184:4): pid=3752 uid=1000 auid=1000 ses=2 subj=unconfined msg='op=PAM:session_close grantors=pam_limits,pam_en
v,pam_env,pam_permit,pam_unix acct="root" exe="/usr/bin/sudo" hostname=kali addr=? terminal=/dev/pts/0 res=success'UID="kali" AUID="kal
i"
type=CRED_DISP msg=audit(1788841332.184:5): pid=3752 uid=1000 auid=1000 ses=2 subj=unconfined msg='op=PAM:setcred grantors=pam_permit acct="root"
exe="/usr/bin/sudo" hostname=kali addr=? terminal=/dev/pts/0 res=success'UID="kali" AUID="kali"
type=CONFIG_CHANGE msg=audit(1788841332.280:6): op=set audit_backlog_limit=8192 old=64 auid=4294967295 ses=4294967295 subj=unconfined res=1AUID="
unset"
type=SYSCALL msg=audit(1788841332.280:6): arch=c000003e syscall=44 success=yes exit=60 a0=3 a1=7fffd47e19e0 a2=3c a3=0 items=0 ppid=3762 pid=3778
auid=4294967295 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=(none) ses=4294967295 comm="auditctl" exe="/usr/sbin/auditctl" subj=
unconfined key=(null)ARCH=x86_64 SYSCALL=sendto AUID="unset" UID="root" GID="root" EUID="root" SUID="root" FSUID="root" EGID="root" SGID="root" F
SGID="root"
type=PROCTITLE msg=audit(1788841332.280:6): proctitle=2F7362696E2F617564697463746C002D52002F6574632F61756469742F61756469742E72756C6573
type=CONFIG_CHANGE msg=audit(1788841332.280:7): op=set audit_failure=1 old=1 auid=4294967295 ses=4294967295 subj=unconfined res=1AUID="unset"
type=SYSCALL msg=audit(1788841332.280:7): arch=c000003e syscall=44 success=yes exit=60 a0=3 a1=7fffd47e19f0 a2=3c a3=0 items=0 ppid=3762 pid=3778
```

```
kali@kali: ~  
$ sudo ausearch -m USER_LOGIN  
<no matches>  
  
$ sudo ausearch -m USER_AUTH  
<no matches>  
  
$ sudo aureport -au  
  
Authentication Report  
=====
```

#	date	time	acct	host	term	exe	success	event
<no events of interest were found>								

```
=====
```

```
$ sudo journalctl -b  
Sep 08 00:19:27 kali kernel: Linux version 6.19.14+kali-amd64 (devel@kali.org) (x86_64-linux-gnu-gcc-15 (Debian 15.2.0-17) 15.2.0, GNU ld (GNU B  
Sep 08 00:19:27 kali kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-6.19.14+kali-amd64 root=UUID=894e89ec-b178-42b7-a450-6e969a2aa8fe ro quiet s  
Sep 08 00:19:27 kali kernel: BIOS-provided physical RAM map:  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000009fbff] usable  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x0000000000009fc00-0x0000000000009ffff] reserved  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x000000000000f0000-0x000000000000ffffff] reserved  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x00000000000100000-0x000000000000dffffff] usable  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x000000000dfff0000-0x00000000dffffff] ACPI data  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x00000000fec00000-0x00000000fec00fff] reserved  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x00000000fee00000-0x00000000fee00fff] reserved  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x00000000fffc0000-0x00000000ffffffff] reserved  
Sep 08 00:19:27 kali kernel: BIOS-e820: [mem 0x00000000100000000-0x0000000011ffffff] usable  
Sep 08 00:19:27 kali kernel: NX (Execute Disable) protection: active  
Sep 08 00:19:27 kali kernel: APIC: Static calls initialized  
Sep 08 00:19:27 kali kernel: SMBIOS 2.5 present.
```

```
kali@kali: ~  
$ journalctl -p err -n 20  
Sep 07 01:21:55 kali lightdm[976]: gkr-pam: unable to locate daemon control file  
Sep 07 01:22:03 kali obexd[1937]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit>  
Sep 07 01:22:03 kali obexd[1937]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit>  
Sep 07 01:25:01 kali CRON[4818]: Authentication token is no longer valid; new one required  
Sep 07 01:35:01 kali CRON[10330]: Authentication token is no longer valid; new one required  
Sep 07 01:39:02 kali CRON[12198]: Authentication token is no longer valid; new one required  
Sep 07 01:45:01 kali CRON[15065]: Authentication token is no longer valid; new one required  
Sep 07 01:55:01 kali CRON[19755]: Authentication token is no longer valid; new one required  
-- Boot 2585e5514e264b248e8ea0c9b793d3bc --  
Sep 08 00:19:27 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.  
Sep 08 00:19:27 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* This configuration is likely broken.  
Sep 08 00:19:27 kali kernel: vmwgfx 0000:00:02.0: [drm] *ERROR* Please switch to a supported graphics device to avoid problems.  
Sep 08 00:19:30 kali kernel: vboxsf: Unknown parameter 'tag'  
Sep 08 00:20:03 kali lightdm[965]: gkr-pam: unable to locate daemon control file  
Sep 08 00:20:12 kali obexd[1973]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit>  
Sep 08 00:20:12 kali obexd[1973]: Unable to acquire registry: Error calling StartServiceByName for org.gnome.evolution.dataserver.Sources5: Unit>  
Sep 08 00:21:30 kali (generato[3339]: '/usr/lib/systemd/system-generators/systemd-sshd-generator' terminated by signal SEGV.  
Sep 08 00:22:25 kali (generato[3935]: '/usr/lib/systemd/system-generators/systemd-sshd-generator' terminated by signal ABRT.  
Sep 08 00:25:01 kali CRON[5259]: Authentication token is no longer valid; new one required  
Sep 08 00:26:17 kali kernel: ohci-pci 0000:00:06.0: frame counter not updating; disabled  
Sep 08 00:26:17 kali kernel: ohci-pci 0000:00:06.0: HC died; cleaning up  
  
$ touch ~/audit_test.txt  
  
$ sudo auditctl -w /home/$USER/audit_test.txt -p rwx -k audit.txt  
Old style watch rules are slower  
  
$ sudo aureport -l  
  
Login Report
```

```
kali@kali: ~  
$ sudo auditctl -l  
-w /home/kali/audit_test.txt -p rwx -k audit.txt  
  
kali@kali: ~  
$ echo "Hello,I am Lunar.I am creating an Audit report for my experiment.Thanks for Listening.Perform it" >> ~/audit_test.txt  
  
kali@kali: ~  
$ chmod 600 ~/audit_test.txt  
  
kali@kali: ~  
$ sudo ausearch -k audit_test.txt  
<no matches>  
  
kali@kali: ~  
$ cat audit_test.txt  
Hello,I am Lunar.I am creating an Audit report for my experiment.Thanks for Listening.Perform it  
  
kali@kali: ~  
$ cat audit_test  
cat: audit_test: No such file or directory  
  
kali@kali: ~  
$ sudo ausearch -k audit_test  
<no matches>  
  
kali@kali: ~  
$ sudo ausearch -k audit.txt  
  
time→Tue Sep  8 00:28:53 2026  
type=PROCTITLE msg=audit(1788841733.657:91): proctitle=617564697463746C002D7002F686F6D652F6B616C692F61756469745F746573742E747874002D700027277861002D6B0061756469742E747874  
type=SYSCALL msg=audit(1788841733.657:91): arch=c000003e syscall=44 success=yes exit=1092 a0=4 a1=7ffff477602f0 a2=444 a3=0 items=0 ppid=7309 pid=7310 auid=1000 uid=0 gid=0 euid=0 suid=0 fsuid=0 egid=0 sgid=0 fsgid=0 tty=pts1 ses=2 comm="auditctl" exe="/usr/sbin/auditctl" subj=unconfined key=(null)  
type=CONFIG_CHANGE msg=audit(1788841733.657:91): auid=1000 ses=2 subj=unconfined op=add_rule key="audit.txt" list=4 res=1  
  
time→Tue Sep  8 00:31:32 2026  
type=PROCTITLE msg=audit(1788841892.772:112): proctitle=63686D6F6400363030002F686F6D652F6B616C692F61756469745F746573742E747874  
type=PATH msg=audit(1788841892.772:112): item=0 name="/home/kali/audit_test.txt" inode=3932814 dev=08:01 mode=0100664 ouid=1000 ogid=1000 rdev=00:00 nametype=NORMAL cap_fp=0 cap_fi=0 cap_fe=0 cap_fver=0 cap_frootid=0  
type=CWD msg=audit(1788841892.772:112): cwd="/home/kali"  
type=SYSCALL msg=audit(1788841892.772:112): arch=c000003e syscall=268 success=yes exit=0 a0=ffffff9c a1=559aaeb3d540 a2=180 a3=fff items=1 ppid=1
```

```
kali@kali: ~  
$ sudo aureport  
  
Summary Report  
Range of time in logs: 09/08/2026 00:22:12.181 - 09/08/2026 00:42:14.883  
Selected time for report: 09/08/2026 00:22:12 - 09/08/2026 00:42:14.883  
Number of changes in configuration: 4  
Number of changes to accounts, groups, or roles: 0  
Number of logins: 0  
Number of failed logins: 0  
Number of authentications: 0  
Number of failed authentications: 0  
Number of users: 2  
Number of terminals: 7  
Number of host names: 2  
Number of executables: 11  
Number of commands: 9  
Number of files: 2  
Number of AVC's: 0  
Number of MAC events: 0  
Number of failed syscalls: 0  
Number of anomaly events: 5  
Number of responses to anomaly events: 0  
Number of crypto events: 0  
Number of integrity events: 0  
Number of virt events: 0  
Number of keys: 1  
Number of process IDs: 29  
Number of events: 166  
  
kali@kali: ~  
$ sudo aureport -au --failed  
  
Authentication Report  
# date time acct host term exe success event  
<no events of interest were found>
```

Conclusion

System logs provide valuable evidence for detecting and investigating suspicious activities. **auditd** provides detailed security auditing, while **journalctl** enables efficient analysis of system and service logs. By correlating events from both sources, security analysts can identify abnormal behavior, investigate potential incidents, and prepare an effective system audit report.